

Assignment 2

MIN-HASHING AND LSH

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Question 1: Create k-Grams

A: Approximate Jaccard similarity between the pair of documents D1 and D2

```
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Exact 3-gram Jaccard(D1, D2) = 0.978

t = 20
Approx Jaccard = 0.9500
Time taken = 0.0054 seconds

t = 60
Approx Jaccard = 0.9833
Time taken = 0.0117 seconds

t = 150
Approx Jaccard = 0.9800
Time taken = 0.0223 seconds

t = 300
Approx Jaccard = 0.9600
Time taken = 0.0330 seconds

t = 600
Approx Jaccard = 0.9883
Time taken = 0.0560 seconds
```

B: Jaccard similarity between all pairs of documents for each type of k-gram

```
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Character 2-gram
D1 - D2 : 0.9811
D1 - D3 : 0.8157
D1 - D4 : 0.6444
D2 - D3 : 0.8000
D2 - D4 : 0.6413
D3 - D4 : 0.6530

Character 3-gram
D1 - D2 : 0.9780
D1 - D3 : 0.5804
D1 - D4 : 0.3051
D2 - D3 : 0.5680
D2 - D4 : 0.3059
D3 - D4 : 0.3121

Word 2-gram
D1 - D2 : 0.9408
D1 - D3 : 0.1823
D1 - D4 : 0.0302
D2 - D3 : 0.1737
D2 - D4 : 0.0303
D3 - D4 : 0.0161
```

Question 2: Min-Hashing

A: Approximate Jaccard similarity between the pair of documents D1 and D2

```
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t = 20
Approximate Jaccard = 0.900000
Time taken = 0.005383 seconds

t = 60
Approximate Jaccard = 0.966667
Time taken = 0.012823 seconds

t = 150
Approximate Jaccard = 0.993333
Time taken = 0.020339 seconds

t = 300
Approximate Jaccard = 0.976667
Time taken = 0.031480 seconds

t = 600
Approximate Jaccard = 0.978333
Time taken = 0.054593 seconds

t = 800
Approximate Jaccard = 0.978750
Time taken = 0.070960 seconds

t = 1000
Approximate Jaccard = 0.975000
Time taken = 0.131359 seconds

t = 1500
Approximate Jaccard = 0.984000
```

B: Analysis on t-value

- Jaccard similarity between the two documents is **0.978**. As we increase the number of hash functions (**t**), the MinHash estimate becomes closer to this exact value.
- For small **t** values (like **t = 20 or 60**), the approximation is less stable, and the error is relatively higher.
- As **t** increases, the error steadily decreases, but the computation time also increases. For example, **t = 600** gives very low error but takes about **0.055 seconds**, which is much slower.
- At **t = 150**, the estimated similarity (**0.9667**) is already very close to the exact value, with a **small error (≈ 0.011)** and reasonable **computation time (~ 0.013 s)**.
- Therefore, **t $\approx$ 150** is a good practical choice, since it provides high accuracy while keeping the computation time efficient.

Question 3: LSH

A: All possible combinations of **b** and **r**

```
LSH Detection Probabilities (for  $\tau = 0.7$ ):  
-----  
r = 1, b = 160  
Estimated Threshold  $\approx 0.0063$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 1.000000  
D1-D4 : 1.000000  
D2-D3 : 1.000000  
D2-D4 : 1.000000  
D3-D4 : 1.000000  
-----  
r = 2, b = 80  
Estimated Threshold  $\approx 0.1118$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 1.000000  
D1-D4 : 0.999597  
D2-D3 : 1.000000  
D2-D4 : 0.999614  
D3-D4 : 0.999725  
-----  
r = 4, b = 40  
Estimated Threshold  $\approx 0.3976$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.991904  
D1-D4 : 0.293930  
D2-D3 : 0.987698  
D2-D4 : 0.296585  
D3-D4 : 0.317129  
-----  
r = 5, b = 32  
Estimated Threshold  $\approx 0.5000$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.886888  
D1-D4 : 0.081202  
D2-D3 : 0.857860  
D2-D4 : 0.082252  
D3-D4 : 0.090568  
-----
```

```
r = 8, b = 20  
Estimated Threshold  $\approx 0.6877$   
>>> Close to  $\tau = 0.7$  <<<  
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.228224  
D1-D4 : 0.001500  
D2-D3 : 0.195880  
D2-D4 : 0.001532  
D3-D4 : 0.001800  
-----  
r = 10, b = 16  
Estimated Threshold  $\approx 0.7579$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.067145  
D1-D4 : 0.000112  
D2-D3 : 0.054526  
D2-D4 : 0.000115  
D3-D4 : 0.000140  
-----  
r = 16, b = 10  
Estimated Threshold  $\approx 0.8660$   
Pair Probabilities:  
D1-D2 : 0.999994  
D1-D3 : 0.001655  
D1-D4 : 0.000000  
D2-D3 : 0.001175  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----  
r = 20, b = 8  
Estimated Threshold  $\approx 0.9013$   
Pair Probabilities:  
D1-D2 : 0.999722  
D1-D3 : 0.000150  
D1-D4 : 0.000000  
D2-D3 : 0.000098  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----  
r = 32, b = 5  
Estimated Threshold  $\approx 0.9509$   
Pair Probabilities:  
D1-D2 : 0.965632  
D1-D3 : 0.000000  
D1-D4 : 0.000000  
D2-D3 : 0.000000  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----  
r = 40, b = 4  
Estimated Threshold  $\approx 0.9659$   
Pair Probabilities:  
D1-D2 : 0.879138  
D1-D3 : 0.000000  
D1-D4 : 0.000000  
D2-D3 : 0.000000  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----
```

```

-----
r = 80, b = 2
Estimated Threshold  $\approx 0.9914$ 
Pair Probabilities:
D1-D2 : 0.308461
D1-D3 : 0.000000
D1-D4 : 0.000000
D2-D3 : 0.000000
D2-D4 : 0.000000
D3-D4 : 0.000000
-----

r = 160, b = 1
Estimated Threshold  $\approx 1.0000$ 
Pair Probabilities:
D1-D2 : 0.028362
D1-D3 : 0.000000
D1-D4 : 0.000000
D2-D3 : 0.000000
D2-D4 : 0.000000
D3-D4 : 0.000000

```

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b	r	b×r	s*	s*-τ
1	160	160	0.0063	0.6937 BEST
2	80	160	0.1118	0.5882 BEST
4	40	160	0.3976	0.3024 BEST
5	32	160	0.5000	0.2000 BEST
8	20	160	0.6877	0.0123 BEST
10	16	160	0.7579	0.0579
16	10	160	0.8660	0.1660
20	8	160	0.9013	0.2013
32	5	160	0.9509	0.2509
40	4	160	0.9659	0.2659
80	2	160	0.9914	0.2914
160	1	160	1.0000	0.3000

Best choice : b = 8 hashes/band, r = 20 bands
 Inflection : s* = 0.6877 (target τ = 0.7)
 S-curve : f(s) = 1 - (1 - s^8)^20

➔ For $\tau = 0.7$ and 'b=80 & r=2' combination, Jaccard similarity between all pairs of documents:

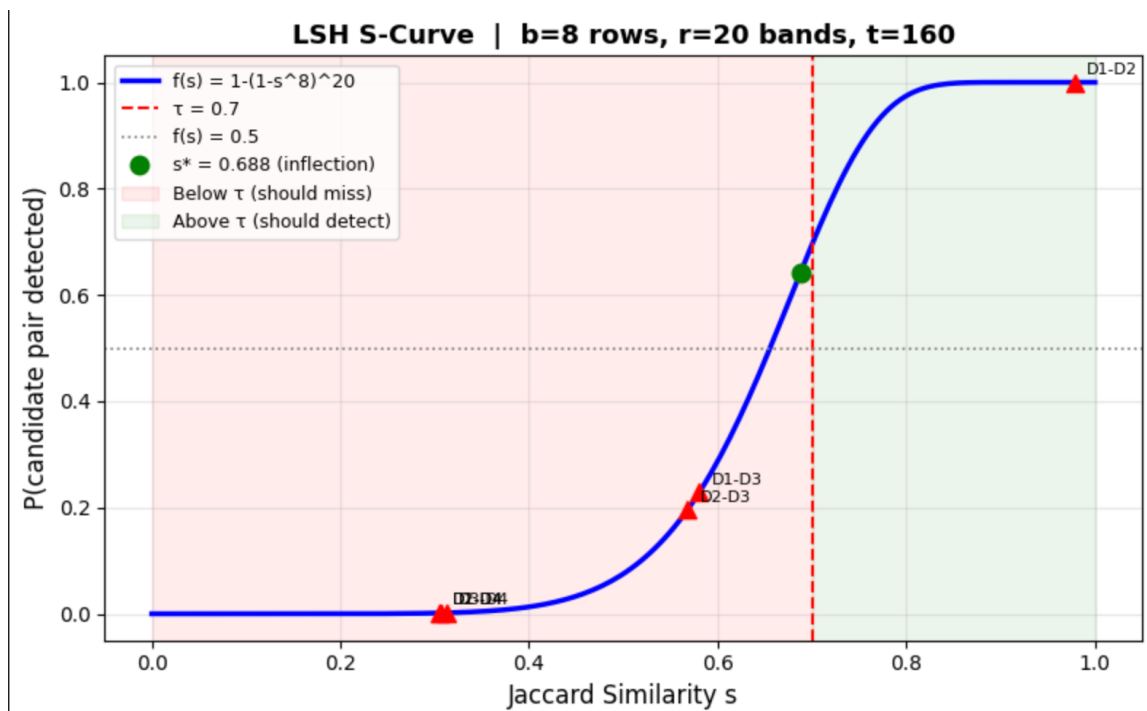
✓ Last Execution: 3:39:46 PM, Duration: 0.0s

Using: b = 8 hashes/band, r = 20 bands, t = 160
 S-curve: f(s) = 1 - (1 - s^8)^20

Pair	Jaccard	f(J) = P(detected)	Above $\tau=0.7$?
D1-D2	0.9780	1.000000	Yes
D1-D3	0.5804	0.228224	No
D1-D4	0.3051	0.001500	No
D2-D3	0.5680	0.195880	No
D2-D4	0.3059	0.001532	No
D3-D4	0.3121	0.001800	No

B: Conclusion (

- **Exact Similarities:**
 - D1–D2 = **0.978** -> clear near-duplicate
 - D1–D3, D2–D3 $\approx$ **0.58** -> moderate similarity
 - Pairs with D4 $\approx$ **0.30** -> low similarity
- **LSH Behaviour:**
 - **r=1, b=160** ($t \approx 0.006$) -> All pairs detected ($P \approx 1$) -> High false positives
 - **r=8, b=20** ($t \approx 0.688 \approx \tau = 0.7$)
 - D1–D2: $P \approx 1$
 - Others: $P \leq 0.23$
-> Best separation
 - **r=160, b=1** ($t \approx 1.0$)
 - D1–D2: $P \approx 0.028$
-> High false negatives
- **Best configuration:** r=8, b=20
- Achieves a good balance between precision and recall
- Proper (r, b) tuning is critical for effective LSH performance



Question 4: Min-Hashing on MovieLens dataset

➔ Load MovieLens dataset

```
Users   : 943
Movies  : 1682
Sample - User 5 rated 175 movies
```

➔ Exact Jaccard for all pairs

```
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Computing exact Jaccard for all 444,153 pairs...
Done in 2.59s
Pairs with  $J \geq 0.5$  : 10

Top 10 most similar pairs:
  User 408 & User 898 →  $J = 0.8387$ 
  User 328 & User 788 →  $J = 0.6730$ 
  User 489 & User 587 →  $J = 0.6299$ 
  User 600 & User 826 →  $J = 0.5455$ 
  User 451 & User 489 →  $J = 0.5333$ 
  User 674 & User 879 →  $J = 0.5217$ 
  User 554 & User 764 →  $J = 0.5170$ 
  User 197 & User 826 →  $J = 0.5130$ 
  User 197 & User 600 →  $J = 0.5000$ 
  User 800 & User 879 →  $J = 0.5000$ 
```

➔ MinHash for $t = [50, 100, 200]$ with 5 runs each

```
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Exact similar pairs (ground truth) : 10

t= 50 | found=6.2 | FP=3.0 | FN=6.8 | time=0.8s
t= 100 | found=3.2 | FP=0.2 | FN=7.0 | time=1.4s
t= 200 | found=2.6 | FP=0.0 | FN=7.4 | time=2.6s
```

➔ For the false positives and negatives, report the averages for 5 different runs.

```
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```

t	Found	Avg FP	Avg FN	FP%	FN%	Time(s)
50	6.2	3.0	6.8	48.4%	68.0%	0.8s
100	3.2	0.2	7.0	6.2%	70.0%	1.4s
200	2.6	0.0	7.4	0.0%	74.0%	2.6s

Question 5: LSH on MovieLens dataset

→ Candidate pairs with a similarity of at least 0.6.

Exact pairs with $J \geq 0.6$: 3

[t=50 r=5 b=10] Run 1 | candidates= 513 | found= 5 | FP= 2 | FN= 0 | 0.2s

[t=50 r=5 b=10] Run 2 | candidates= 599 | found= 5 | FP= 2 | FN= 0 | 0.2s

[t=50 r=5 b=10] Run 3 | candidates= 627 | found= 4 | FP= 2 | FN= 1 | 0.2s

[t=50 r=5 b=10] Run 4 | candidates= 732 | found= 6 | FP= 3 | FN= 0 | 0.2s

[t=50 r=5 b=10] Run 5 | candidates= 469 | found= 3 | FP= 1 | FN= 1 | 0.2s

— AVG: FP=2.0 | FN=0.4 | Found=4.6

[t=100 r=5 b=20] Run 1 | candidates=1348 | found= 2 | FP= 1 | FN= 2 | 0.4s

[t=100 r=5 b=20] Run 2 | candidates= 901 | found= 2 | FP= 0 | FN= 1 | 0.4s

[t=100 r=5 b=20] Run 3 | candidates= 978 | found= 4 | FP= 1 | FN= 0 | 0.4s

[t=100 r=5 b=20] Run 4 | candidates= 945 | found= 3 | FP= 0 | FN= 0 | 0.4s

[t=100 r=5 b=20] Run 5 | candidates=1028 | found= 4 | FP= 1 | FN= 0 | 0.4s

— AVG: FP=0.6 | FN=0.6 | Found=3.0

[t=200 r=5 b=40] Run 1 | candidates=2449 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 2 | candidates=2202 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 3 | candidates=1803 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 4 | candidates=2496 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 5 | candidates=1707 | found= 3 | FP= 0 | FN= 0 | 0.8s

— AVG: FP=0.0 | FN=0.0 | Found=3.0

[t=200 r=10 b=20] Run 1 | candidates= 6 | found= 1 | FP= 0 | FN= 2 | 0.8s

[t=200 r=10 b=20] Run 2 | candidates= 8 | found= 1 | FP= 0 | FN= 2 | 0.8s

[t=200 r=10 b=20] Run 3 | candidates= 2 | found= 1 | FP= 0 | FN= 2 | 0.7s

[t=200 r=10 b=20] Run 4 | candidates= 4 | found= 1 | FP= 0 | FN= 2 | 0.7s

[t=200 r=10 b=20] Run 5 | candidates= 6 | found= 2 | FP= 0 | FN= 1 | 0.7s

— AVG: FP=0.0 | FN=1.8 | Found=1.2

→ The number of false positives and false negatives, taking the average over 5 runs.

$\tau = 0.6$ Exact pairs = 3						
Config	Found	Avg FP	Avg FN	FP%	FN%	Time
t=50 r=5 b=10	4.6	2.0	0.4	43.5%	13.3%	0.2s
t=100 r=5 b=20	3.0	0.6	0.6	20.0%	20.0%	0.4s
t=200 r=5 b=40	3.0	0.0	0.0	0.0%	0.0%	0.8s
t=200 r=10 b=20	1.2	0.0	1.8	0.0%	60.0%	0.8s

➔ Candidate pairs with a similarity of at least 0.8.

Exact pairs with $J \geq 0.8$: 1

[t=50 r=5 b=10] Run 1 | candidates= 574 | found= 1 | FP= 0 | FN= 0 | 0.2s
[t=50 r=5 b=10] Run 2 | candidates= 347 | found= 1 | FP= 0 | FN= 0 | 0.2s
[t=50 r=5 b=10] Run 3 | candidates= 604 | found= 1 | FP= 0 | FN= 0 | 0.2s
[t=50 r=5 b=10] Run 4 | candidates= 492 | found= 1 | FP= 0 | FN= 0 | 0.2s
[t=50 r=5 b=10] Run 5 | candidates= 453 | found= 1 | FP= 0 | FN= 0 | 0.2s
— AVG: FP=0.0 | FN=0.0 | Found=1.0

[t=100 r=5 b=20] Run 1 | candidates= 876 | found= 0 | FP= 0 | FN= 1 | 0.4s
[t=100 r=5 b=20] Run 2 | candidates= 760 | found= 0 | FP= 0 | FN= 1 | 0.4s
[t=100 r=5 b=20] Run 3 | candidates=1273 | found= 1 | FP= 0 | FN= 0 | 0.4s
[t=100 r=5 b=20] Run 4 | candidates=1878 | found= 1 | FP= 0 | FN= 0 | 0.4s
[t=100 r=5 b=20] Run 5 | candidates=1343 | found= 1 | FP= 0 | FN= 0 | 0.4s
— AVG: FP=0.0 | FN=0.4 | Found=0.6

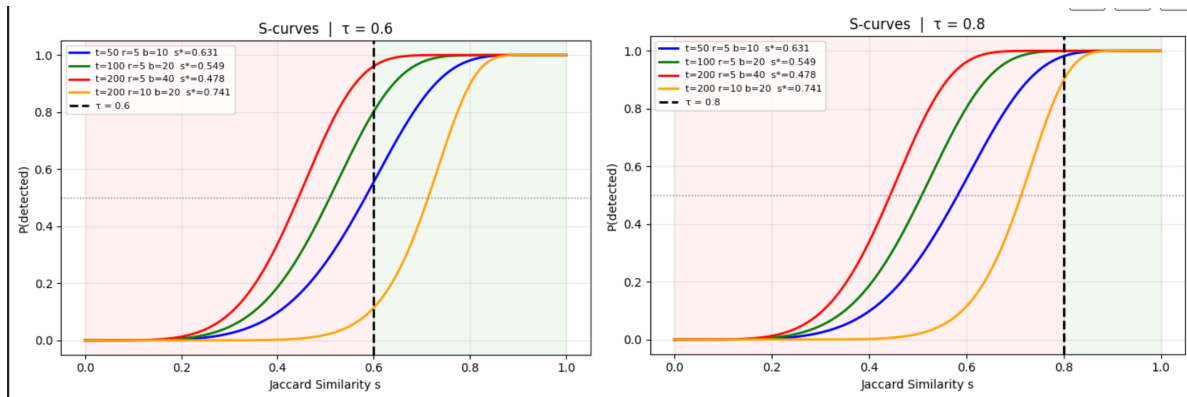
[t=200 r=5 b=40] Run 1 | candidates=1993 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=5 b=40] Run 2 | candidates=1736 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=5 b=40] Run 3 | candidates=2554 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=5 b=40] Run 4 | candidates=1978 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=5 b=40] Run 5 | candidates=2544 | found= 1 | FP= 0 | FN= 0 | 0.8s
— AVG: FP=0.0 | FN=0.0 | Found=1.0

[t=200 r=10 b=20] Run 1 | candidates= 12 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=10 b=20] Run 2 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=10 b=20] Run 3 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=10 b=20] Run 4 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s
[t=200 r=10 b=20] Run 5 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s
— AVG: FP=0.0 | FN=0.0 | Found=1.0

➔ The number of false positives and false negatives, taking the average over 5 runs.

$\tau = 0.8$ Exact pairs = 1						
Config	Found	Avg FP	Avg FN	FP%	FN%	Time
t=50 r=5 b=10	1.0	0.0	0.0	0.0%	0.0%	0.2s
t=100 r=5 b=20	0.6	0.0	0.4	0.0%	40.0%	0.4s
t=200 r=5 b=40	1.0	0.0	0.0	0.0%	0.0%	0.8s
t=200 r=10 b=20	1.0	0.0	0.0	0.0%	0.0%	0.8s

→ LSH S-curves for all configs



→ Analysis

- $\tau = 0.6$: Using more hash functions generally improved accuracy. The configuration **($t=200, r=5, b=40$)** performed best with **0% false positives and 0% false negatives**.
- The combination **($t=50, r=5, b=10$)** produced many **false positives (~43.5%)**, meaning it detected many pairs that were not actually similar.
- **($t=200, r=10, b=20$)** reduced false positives but introduced **high false negatives**, missing several truly similar pairs.
- $\tau = 0.8$: Since the similarity requirement is stricter, fewer true pairs exist. Most configurations had **0% false positives**, but **($t=100, r=5, b=20$)** missed some valid pairs (**~40% false negatives**).
- Overall, higher hash counts with balanced banding (e.g., **$t=200, r=5, b=40$**) gave the most reliable results.